Contents

1 Chats

Exported on: 1/24/2026, 9:16:29 AM

1.1 👤 You

solve integral and describe solution step by step

Please check with https://math.stackexchange.com/questions/562694/integral-int-11-frac1x-sqrt-frac1x1-x-ln-left-frac2-x22-x1/563063#563063

1.2 🤖 Gemini

Show thinking

Based on the verification with the provided Math StackExchange link, specifically the answer by Ron Gordon (and confirmed by others like Ivan and Cleo), here is the step-by-step solution to the integral.

The integral is:

$$I = \int_{-1}^{1} \frac{1}{x}\sqrt{\frac{1+x}{1-x}} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right) dx$$

Step 1: Algebraic Substitution

First, simplify the square root term using the substitution $t = \frac{1-x}{1+x}$. This maps the interval $(-1, 1)$ to $(0, \infty)$. This transforms the integral into a form involving a rational function and a logarithm. After applying this and subsequent substitutions (specifically $t = u^2$ to remove the square root, and then $u = \frac{v-1}{v+1}$), the integral can be converted into a form suitable for advanced techniques.

A key intermediate form derived in the linked solution is:

$$I = 8\int_0^{\infty} \frac{(v^2-1)(v^4-6v^2+1)}{v^8+4v^6+70v^4+4v^2+1} \ln(v)\, dv$$

Step 2: Complex Analysis (Contour Integration)

The resulting integral is a rational function times a logarithm over the positive real line. This is a classic candidate for Contour Integration using the Residue Theorem.

Consider a keyhole contour in the complex plane.

The integral relates to the sum of residues of the function $f(z) = \frac{P(z)}{Q(z)}(\ln z)^2$ at its poles, where $Q(z)$ is the 8th-degree polynomial in the denominator.

The roots of the denominator exploit the symmetry of the Golden Ratio, $\phi = \frac{1+\sqrt{5}}{2}$. The poles are located at complex points related to ϕ.

Step 3: Summing Residues

Calculating the residues at the 8 poles and summing them (exploiting symmetries like $z \to 1/z$ and $z \to -z$) yields a surprisingly clean result involving the arccotangent of the square root of the Golden Ratio.

Final Solution

The exact closed-form value of the integral is:

$$I = 4\pi \operatorname{arccot}\left(\sqrt{\phi}\right)$$

where $\phi = \frac{1+\sqrt{5}}{2}$ is the Golden Ratio.

Alternatively, this can be expressed using other inverse trigonometric functions:

$$I = 4\pi \arcsin\left(\frac{1}{\phi}\right) = 4\pi \arcsin\left(\frac{\sqrt{5}-1}{2}\right)$$

$$I = \pi \arccos\left(17 - 8\sqrt{5}\right)$$

Numerical Value:

$$I \approx 8.3722116\ldots$$
